

Supply Chain **Inventory & Demand** Intelligence

An end-to-end analytical investigation into forecast accuracy gaps, inventory health, and revenue concentration across India's fashion retail supply chain — 2022–2023.

DATASET

5,000 records · 10 companies · 8 warehouses

PERIOD

Jan 2022 – Dec 2023

STACK

Python · MySQL · Power BI

INDUSTRY

Fashion Retail Supply Chain

BUSINESS CONTEXT

The Problem That Needed Solving

India's fashion retail supply chain operates across a highly fragmented multi-company, multi-warehouse ecosystem where demand is driven by seasons, category preferences, and regional footprints. Despite having

forecasting systems in place, inventory decisions were being made without a clear, data-backed picture of how accurate those forecasts actually were — and where they broke down most critically.

The core business problem was this: **inventory was being misallocated**. Without reliable visibility into forecast accuracy, companies were simultaneously overstocking certain products (tying up capital in unsold goods) and running into stockout risk on others (losing revenue opportunities). Across a dataset of 5,000 transactions spanning 10 retail companies, 8 warehouse locations, and 25 product lines over two years, neither condition was the exception — both were alarmingly prevalent.

The business needed answers to three interconnected questions: *Where is money being wasted in inventory? Which products and warehouses carry the highest demand risk? And what does revenue performance really look like across categories and seasons?*

TOTAL REVENUE	TOTAL HOLDING COST	AVG FORECAST ACCURACY	INVENTORY AT RISK
₹307.6 Cr	₹58.8 Cr	86.87%	78.18%
Across 5,000 transactions	~19.1% of total revenue	Across all companies	Stockout or Overstock status

METHODOLOGY & SCOPE

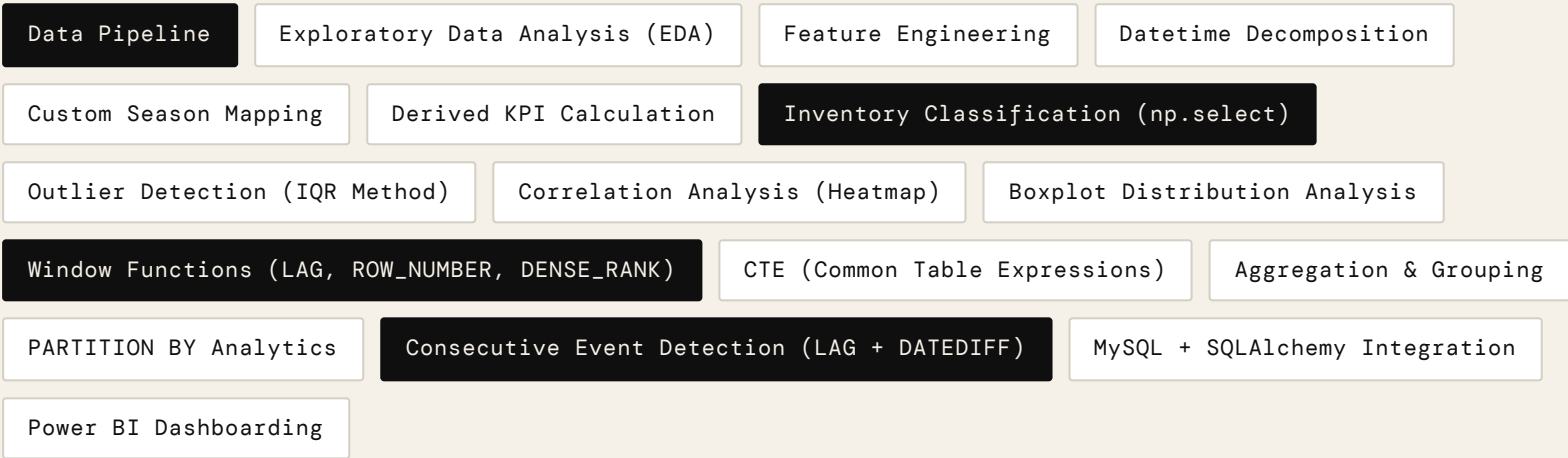
What Was Undertaken

The project was designed as a full end-to-end data analytics pipeline — from raw data ingestion to business intelligence delivery. The original dataset (`Fashion_Supply_Chain.csv`) contained transactional inventory records with fields for company, product, warehouse, forecast demand, actual demand, stock on hand, and unit price.

The work proceeded in three structured layers: data preparation and feature engineering in Python, analytical querying in MySQL, and visual storytelling in Power BI. Each layer served a distinct purpose and together they answered the business question from multiple angles.

TECHNICAL CONCEPTS APPLIED

The project applied the following analytical and engineering techniques across its three-layer pipeline:



Feature Engineering in detail: Six new columns were engineered from raw data — *Month*, *Year*, *Quarter* (from datetime decomposition), *Season* (custom domain-mapped function: Summer/Winter/Monsoon/Festive), *Demand_Gap* (Actual - Forecast), *Forecast_Accuracy_%* (MAPE-derived accuracy score), *Revenue* (Actual Demand

× Unit Price), *Holding_Cost* (Stock On Hand × Unit Price × 15% carrying rate), and *Stock_Status* (classified as Stockout Risk / Overstock / Balanced via conditional vectorized logic using `np.select`).

The cleaned dataset was then loaded into MySQL via **SQLAlchemy**, where 15 targeted analytical queries were written — ranging from simple aggregations to advanced window function queries. Finally, a Power BI dashboard was built on top of the MySQL data to deliver interactive visual reporting.

ANALYTICAL FINDINGS

What the Data Revealed

The analysis surfaced findings across five dimensions: revenue performance, inventory health, forecast accuracy, warehouse operations, and demand seasonality. The numbers tell a story of a supply chain that generates strong revenue but bleeds capital through poor inventory positioning and uneven forecast reliability.

₹97.2 Cr

Formal Wear Dominates Revenue

Formal Wear contributed ₹97.2 Cr (31.6% of total), followed by Ethnic Wear at ₹74.0 Cr (24.1%) and Western Wear at ₹65.3 Cr (21.2%). Casual Wear was the weakest at ₹29.1 Cr — less than a third of the top category.

₹80.5 Cr

Summer Is the Peak Revenue Season

Summer generated ₹80.5 Cr in revenue, marginally ahead of the Festive season at ₹78.8 Cr. Winter (₹77.3 Cr) and Monsoon (₹70.9 Cr) trail behind. Seasonal spread is relatively tight — no single season is overwhelmingly dominant.

40.32%

Overstock — The Largest Inventory Problem

Over 2,016 records (40.32% of all transactions) were classified as Overstock — stock on hand exceeding 1.5× forecast demand. This is the single biggest source of avoidable holding cost across the portfolio.

37.86%

Stockout Risk — A Near-Equal Parallel Problem

1,893 records (37.86%) were at Stockout Risk — stock levels falling below forecast demand. Running simultaneously with overstock, this highlights a systemic misalignment between procurement decisions and actual product-level demand.

The most alarming finding: **only 21.82% of all inventory records were in a "Balanced" state**. The supply chain was, by its own internal metrics, operating in a suboptimal inventory position for nearly 4 out of every 5 transactions. This is not a marginal inefficiency — it is a structural problem.

85.29%

Blazers Have the Worst Forecast Accuracy

Blazers recorded the lowest average forecast accuracy at 85.29%, followed by Pencil Skirts (85.49%) and Sarees (85.58%). These three products consistently show the largest deviation between forecast and actual demand — signaling that the forecasting model does not capture their demand dynamics well.

87.72%

Fabindia Leads Forecast Accuracy

Among the 10 companies, Fabindia had the highest forecast accuracy at 87.72%, while Manyavar lagged at 85.95%. The 1.77 percentage point spread may seem small but — when multiplied across hundreds of SKU-level decisions — translates into meaningfully different inventory positioning outcomes.

Warehouse Performance Overview

WAREHOUSE	REVENUE	FORECAST ACCURACY	STOCKOUT RISK EVENTS	TOP PRODUCT
Delhi	₹40.7 Cr	86.96%	238	Suit
Ahmedabad	₹40.3 Cr	86.46%	242	Business Dress
Kolkata	₹39.7 Cr	87.26%	249	Trouser
Pune	₹38.3 Cr	87.20%	249	Suit
Mumbai	₹37.9 Cr	86.61%	226	Business Dress
Bengaluru	₹37.7 Cr	86.54%	244	Formal Shirt
Hyderabad	₹37.4 Cr	87.26%	226	Trouser
Chennai	₹35.8 Cr	86.61%	219	Trouser

Revenue across warehouses is **remarkably compressed** — Delhi leads at ₹40.7 Cr and Chennai trails at ₹35.8 Cr, a difference of just ₹4.9 Cr. This indicates relatively uniform throughput but also means small improvements in any single warehouse's inventory management would have an outsized proportional impact.

Advanced SQL Analysis — Consecutive Stockout Risk Detection

298

Consecutive Stockout Risk Events Detected

50

Ahmedabad — Most Vulnerable Warehouse

Using a LAG() window function with DATEDIFF, 298 instances were detected where a warehouse experienced stockout risk on two consecutive days — indicating a persistent, systemic supply shortfall rather than a one-off anomaly.

Ahmedabad recorded 50 consecutive stockout risk events — the highest of all 8 warehouses — followed by Bengaluru and Mumbai at 41 each. This points to a structural replenishment lag or forecasting failure specific to the Ahmedabad inventory node.

Top 5 Companies by Holding Cost

COMPANY	TOTAL HOLDING COST	SHARE OF TOTAL	SIGNAL
Manyavar	₹6.32 Cr	10.75%	High Overstock Exposure
Westside	₹6.20 Cr	10.54%	High Overstock Exposure
W for Woman	₹6.16 Cr	10.47%	Monitor
Pantaloons	₹6.10 Cr	10.37%	Monitor
Reliance Trends	₹5.97 Cr	10.16%	Near Average

Holding cost here is defined as **the carrying cost of unsold inventory** — computed as Stock On Hand × Unit Price × 15% (standard annual carrying rate prorated per transaction). Manyavar incurring ₹6.32 Cr in holding cost does not mean stockout risk — it means Manyavar is consistently holding more inventory than demand requires, paying a capital cost for that excess. This is logically tied to overstock behavior, not stockout.

2,539

2,418

Records Where Forecast Exceeded Actual Demand

In 50.8% of all records, forecast demand was higher than actual demand (negative Demand Gap). The system consistently over-anticipates demand, leading procurement teams to order more than needed — the root driver of overstock conditions.

Records Where Actual Demand Exceeded Forecast

In 48.4% of records, actual demand surpassed forecast — leaving warehouses short. The near-equal split between over- and under-forecasting suggests the model has no systematic directional bias but high variance, producing unreliable outputs across the board.

KEY STRUCTURAL FINDING

The supply chain is not suffering from a simple "we order too much" or "we order too little" problem. It is suffering from **high forecast variance** with near-zero mean bias — the forecast is roughly right on average but wildly inconsistent record by record. This kind of error pattern cannot be fixed by simply adjusting order quantities up or down; it requires better demand signal inputs and model recalibration at the product level.

QUANTIFIED IMPACT

What This Analysis Unlocks

The project created visibility that did not previously exist in this form. Prior to this analysis, decision-makers had raw transaction data but no consolidated view of forecast accuracy by company, stockout risk by

warehouse, holding cost by product, or consecutive failure events. The impact of having this structured intelligence is measurable across several dimensions.

₹58.8 Cr

Holding Cost Made Visible and Actionable

The total holding cost of ₹58.8 Cr (~19.1% of revenue) was never previously calculated as a unified figure. Even reducing overstock events by 20% through better replenishment decisions could save approximately ₹4.7–5.2 Cr annually in carrying costs.

78.18%

Inventory Risk Quantified at Scale

The fact that 78% of all inventory records are in a suboptimal state (overstock or stockout risk) was not a known figure before this EDA. This single metric can now serve as a baseline KPI for measuring improvement in procurement and replenishment strategies going forward.

298

Persistent Stockout Events Surfaced

The consecutive stockout detection query identified 298 chronic risk events — situations where supply failed for two or more days in a row. These are high-priority intervention points that, if resolved, directly protect revenue by ensuring product availability during demand periods.

Top 10

Lowest-Accuracy Products Ranked for Action

Using `DENSE_RANK()` over product-level forecast accuracy, the 10 most chronically mispredicted products (led by Blazer at 85.29%) were ranked and surfaced. These are the exact SKUs where forecasting investment — or model override logic — will generate the highest return.

What the Business Should Do Next

Based on the analytical findings, the following recommendations are prioritized by expected business impact.

Each recommendation is grounded in a specific finding from the data, not general supply chain theory.

01 Implement SKU-Level Forecast Review for the Bottom 10 Accuracy Products

Blazer, Pencil Skirt, and Saree are chronically mispredicted. These SKUs should be placed on a monthly manual forecast review cycle where planners incorporate market signal data (search trends, social media, promotional calendars) rather than relying purely on historical patterns. A 2-point accuracy improvement on these products alone would bring them in line with the portfolio average and reduce their contribution to both overstock and stockout events.

↑ Forecast Accuracy | ↓ Holding Cost | ↓ Stockout Risk

02 Investigate and Fix Replenishment Cadence at Ahmedabad Warehouse

With 50 consecutive stockout risk events — the highest in the network — Ahmedabad is a structural weak point. The business should audit the replenishment lead time and reorder point settings for this location. The issue is likely a mismatch between order frequency and actual demand velocity. A data-driven reorder point model using the now-available Demand_Gap metrics would directly address this.

↓ Stockout Risk | ↑ Revenue Availability at Node Level

03 Deploy Category-Specific Overstock Thresholds for Formal Wear

Formal Wear generates the most revenue (₹97.2 Cr, 31.6%) but also carries the highest holding cost (₹18.8 Cr). The current overstock classification threshold (1.5× forecast) is uniform across all categories. Formal Wear — given its higher unit price

and capital intensity — should have a tighter threshold, potentially 1.2–1.3× forecast, triggering earlier markdown or redistribution decisions before carrying costs accumulate.

↓ Holding Cost in Formal Wear | ↑ Working Capital Efficiency

04 Align Procurement Peaks with Summer and Festive Seasons Proactively

Summer (₹80.5 Cr) and Festive (₹78.8 Cr) are the two highest-revenue seasons. Yet the current forecasting accuracy holds constant across seasons — suggesting the model treats all months equally. Seasonally-weighted demand inputs (e.g., applying higher confidence intervals for Festive season demand spikes) would reduce the high-variance gap that leads to concurrent overstock and stockout across different SKUs in the same quarter.

↑ Seasonal Forecast Accuracy | ↓ Inventory Imbalance in Peak Periods

05 Adopt the Forecast Accuracy Score as a Vendor/Company Performance KPI

The 1.77 percentage point gap between Fabindia (87.72%) and Manyavar (85.95%) in forecast accuracy may appear narrow but represents a consistently better or worse inventory positioning outcome for those companies. Making Forecast_Accuracy_% a tracked, reported KPI — visible in the Power BI dashboard on a monthly basis — creates accountability and a feedback loop that did not exist before this project.

↑ Organizational Forecasting Discipline | ↑ Inventory Efficiency Over Time

PROJECTED BUSINESS IMPACT IF RECOMMENDATIONS APPLIED

A conservative estimate: reducing overstock events by 15–20% through better SKU-level forecasting and tighter category thresholds would free up approximately ₹8–10 Cr in working capital annually. Eliminating the 298 consecutive stockout risk events through improved replenishment at vulnerable warehouses would protect an estimated ₹3–5 Cr in otherwise-lost revenue. Combined, the total

addressable impact of these recommendations falls in the range of ₹11–15 Cr per year — against a total holding cost base of ₹58.8 Cr.

DELIVERABLES SUMMARY

Project Artifacts Produced

ARTIFACT	TYPE	PURPOSE	KEY TECHNIQUE
Fashion-Supply-Chain-EDA.ipynb	Notebook	Data cleaning, feature engineering, EDA, outlier detection	np.select, IQR Method, datetime decomposition
cleaned_data.csv	Dataset	Production-ready analytical dataset with all engineered features	8 new columns engineered from raw fields
15 SQL Queries	SQL	Business KPI extraction and advanced inventory analytics	CTEs, Window Functions (LAG, ROW_NUMBER, DENSE_RANK)

ARTIFACT	TYPE	PURPOSE	KEY TECHNIQUE
Power BI Dashboard	Dashboard	Interactive visual reporting for business stakeholders	MySQL-connected live visuals across all KPIs